

ALEJANDRA RINCÓN

SENIOR DATA SCIENTIST

Senior Data Scientist with a PhD in Mathematics and a proven track record of developing end-to-end data solutions to drive measurable impact. Leveraged Gen AI, advanced machine learning techniques, deep statistical modeling, and big-data architectures to uncover actionable insights. Skilled in interdisciplinary collaboration, leading data-driven initiatives for scalable business outcomes. Passionate about connecting research with industry applications to deliver transformative results.

CONTACT INFO

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 **arinconh**

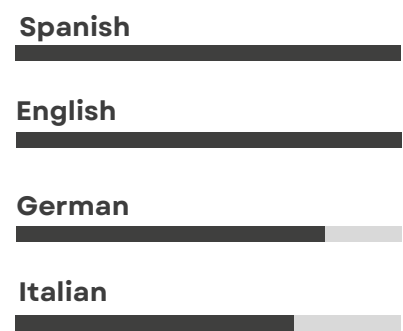
 **arinconh**

 **Berlin, Germany**

TECHNICAL SKILLS

- **Python**
- **SQL**
- **Machine Learning (Pytorch, sklearn)**
- **Gen AI (MCP, LangGraph, LangSmith, Ollama)**
- **Fast API**
- **Airflow**
- **PyMC**
- **HPC**
- **AWS (Bedrock, SageMaker, S3, QuickSight)**

LANGUAGES



WORK EXPERIENCE

Data Scientist 2023-
NET CHECK

- Built an agentic AI MVP for network planning optimization with React frontend and FastAPI backend, deployed on AWS. Implemented RAG to query unstructured documents (PDFs, regulatory docs). Integrated Box, GitHub, and Confluence from the company via Model Context Protocol (MCP). It was developed using LangGraph to automate technical report generation, reducing manual workflow time.
- Analyzed contact networks using crowdsourced GPS data representing 1% of the German population to uncover human interaction patterns. Developed advanced models integrating contact networks and Bayesian Marketing Mix Models to forecast epidemiological trends. Using contact patterns as a benchmark, we computed the increased transmissibility of emerging variants, revealing Delta and Omicron to be 50% and 115% more transmissible than Alpha, respectively. Presented this work at international conferences.
- Implemented the Deep Modularity Network (DMoN) for clustering and community detection in Germany for segregation studies.
- Built and maintained a robust data processing pipeline with Apache Airflow, ensuring scalability and efficiency.

Postdoctoral Researcher 2019-2022
International Center for Theoretical Physics

- Authored scientific papers in algebraic geometry for publishing in peer-reviewed journals.
- Collaborated with researchers worldwide and gave research talks to communicate my current research and its impact effectively.

Teaching Assistant 2014-2022
Freie Universität Berlin
International Center for Theoretical Physics

- Motivated students to be actively engaged throughout the learning process, helped them to create a more diverse, engaged and active math community.

EDUCATION

Ph.D in Mathematics (summa cum laude) 2014 - 2019
Freie Universität Berlin
 Berlin, Germany

M.Sc in Mathematics 2011 - 2014
Bonn Universität
 Bonn, Germany

B.Sc in Mathematics (with honors) 2005 - 2010
Universidad Nacional de Colombia
 Bogotá, Colombia